

CAPSTONE — EXECUTIVE SUMMARY

Public-Data Analytics on the InGen Robotics Portfolio

Market · Competition · BI · Forecasting · Operations

Thirteen weeks, five verticals, **public data only** — no internal access required.

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Six findings that matter

If you read one slide, read this one.

1

Funding is not a moat

Capital raised and defensible IP diverge. Margin structure predicts durability — not headline raises.

2

The biggest market isn't the loudest

Sentinel (indoor security) has the clearest near-term economics. Humanoid gets the capital and the headlines.

3

Attention ≠ opportunity

Demand ranking is the near-inverse of market attractiveness. Sentinel: strong asset, weak share of voice.

4

Momentum is up; models beat naive

MASE < 1.0 in 4 of 5 verticals. No single model wins everywhere — keep a portfolio.

5

Sentinel's alerts: persistence, not threshold

120-min persistence cuts alert load 33× (16.35 → 0.50/day) while recall rises to 88%.

6

Support is a parts problem

Parts = 53% of cycle time, 100% waiting. Parts buffer -23% vs +1 engineer -8%.

How this was built

Four phases, thirteen self-contained modules, everything reproducible.

Phase 1

Wks 1-3

Foundation

Products, 40+ competitors, 12-dataset pipeline, DuckDB warehouse



Phase 2

Wks 4-6

Market analytics

TAM/SAM/SOM, demand-signal index, peer financials



Phase 3

Wks 7-9

BI platform

Star schema, 3 dashboards, exec scorecard



Phase 4

Wks 10-12

Advanced analytics

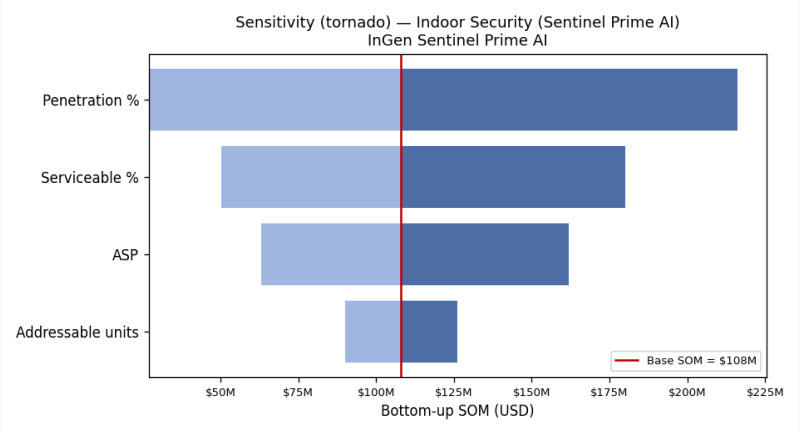
Forecasting, anomaly detection, process optimisation

Principles: sourced or not stated · ranges over false precision · synthetic data always labelled · fixed seeds + test suites · methods validated against ground truth where it exists

Market sizing — ranges, not points

Published estimates disagree by an order of magnitude. That's the finding.

Sentinel Prime AI indoor security	Highest	Guard-replacement economics vs a costed alternative
Fari eldercare	Medium-high	Large population; adoption-rate constrained
Senpai education	Medium	Steady; tied to budget cycles
Aido Rover outdoor patrol	Scenario range	High potential, high uncertainty
Aido Humanoid humanoid	Scenario range	A bet on future deployment, not a current pool



Penetration rate dominates

In every vertical: penetration first, ASP second, unit count a distant third. Market size is mostly a go-to-market variable InGen partly controls — not a fixed external quantity.

Humanoid quoted ~\$290M–\$4.89bn; security robotics ~\$4.7bn–\$19bn. Every figure is a sourced range.

Competition — funding is not a moat

Weeks 1-2

15 priority peers, real patents and headcounts, dated sources.

Capital raised and defensible IP diverge sharply.

The most heavily funded humanoid companies are not the most patent-dense. Several quieter competitors hold stronger, earlier, more specific grants. Funding is the metric the trade press reports and the one that shapes internal anxiety — and it is the wrong one.

\$1bn + thin patents

Buying time

A war chest funds runway, not defensibility.

\$100M + dense position

Buying ground

Specific grants around a mechanism are durable.

Confirmed in Week 6

From the other side

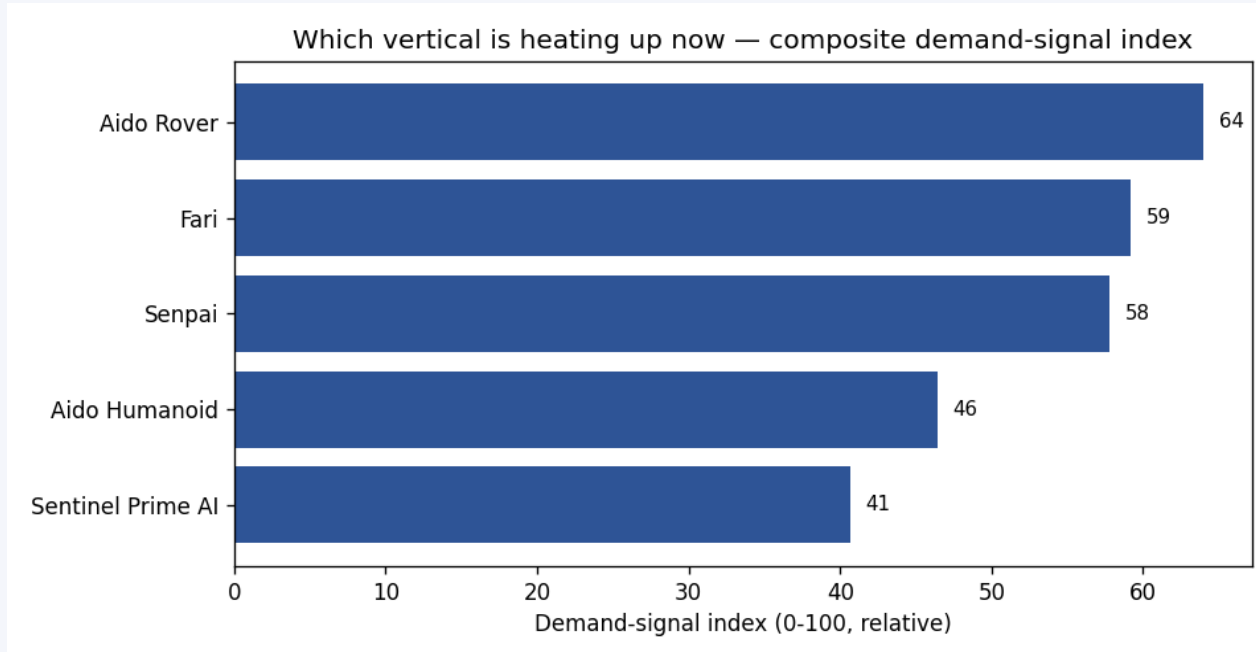
Durable peers have defensible margin structures — not the biggest raises.

Excluded on purpose: real-time hiring counts (no dated snapshot) and vendor deployment claims sourced only to marketing.

Attention $\neq$ opportunity

Week 5

The demand ranking is the near-inverse of market attractiveness.



The inversion

Aido Rover leads on attention (64). Sentinel Prime AI — the clearest near-term economics — comes **last (41)**.

These aren't contradictory. Search and news measure attention. Sizing measures money. The gap is where the commercial opportunity hides.

Sentinel = strong asset, weak share of voice. A marketing problem — solvable.

Two pain-points recur in every vertical: navigation reliability in cluttered real spaces, and the gap between demo behaviour and sustained field behaviour. Both are credibility problems — both answered by published field-reliability data.

Financial benchmarks — margin structure is the moat

Real FY2024 filings. The profitable peers aren't 'robotics companies' as the press means it.

Company	FY24 revenue	Gross margin	Op margin	What it tells us
Teradyne	\$2,820M	58.5%	+20.5%	Test & measurement — the profile robotics aspires to
Cognex	\$915M	68.0%	+7.0%	Machine vision — highest margin; a components business
Symbotic	\$1,790M	18.0%	-5.0%	Warehouse automation — scale, thin margin, integration economics
iRobot	\$681.8M	20.9%	-15.1%	Consumer robotics pure-play — losing money at scale

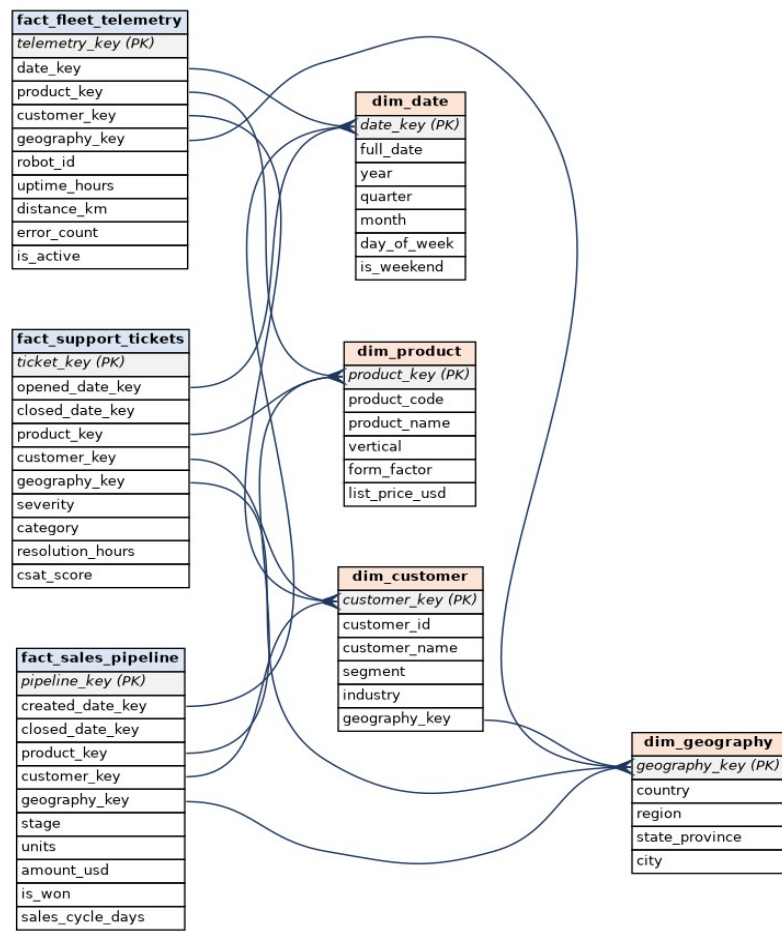
Figure AI's ~\$1.68bn raise is ~2.5× iRobot's entire FY2024 revenue.

Capital is not the constraint in this industry — and it is not the moat either. What separates durable from fragile is whether you own something specific enough to charge for. For InGen: Sentinel-style deployments sold against a costed guard alternative have a defensible price story. Volume consumer hardware, on this evidence, does not.

The BI platform

Weeks 7-9

A warehouse an analytics team could actually query, and dashboards on top.



Star schema

3 facts (telemetry, tickets, pipeline) + 4 dimensions. Surrogate keys, documented grain, zero orphan FKs on every load.

100k rows, fixed seed

Synthetic — no internal data was available. It fixes the shape, not the facts.

Portable by design

The schema mirrors what an InGen team would hold, so the 15-query library and every dashboard move to real data unchanged.

3 tools, 3 jobs

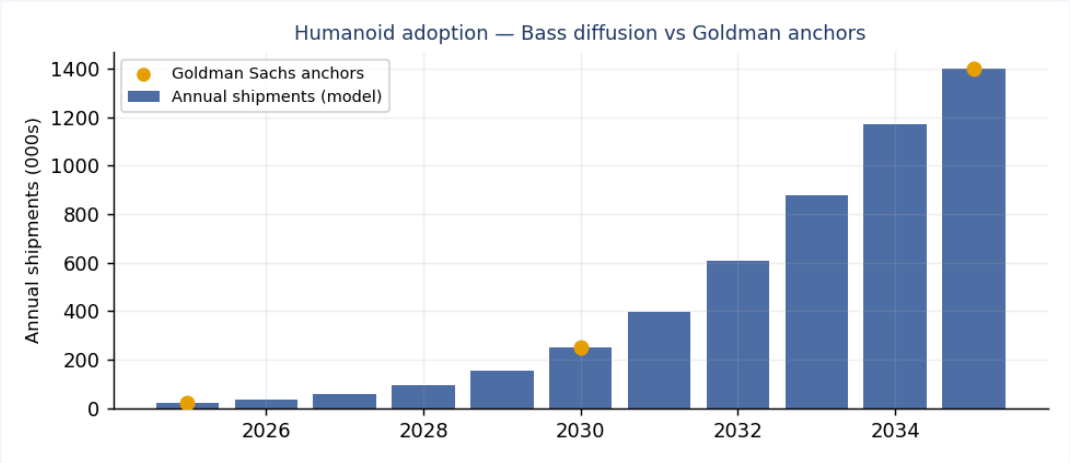
Tableau (external market view) · Looker (operations) · Power BI (exec scorecard, 10 DAX measures).

Dashboards are built and specified; publishing to the live services runs through personal accounts and was not completed. No live URLs are claimed.

Forecasts — models beat naive in 4 of 5

5 models per vertical, backtested on a held-out year. MASE < 1.0 beats seasonal-naive.

Vertical	Best model	MAPE	MASE
Senpai — education	Prophet	3.1%	0.54
Aido Humanoid	ETS	2.6%	0.59
Aido Rover — patrol	Prophet	2.5%	0.69
Sentinel — security	XGBoost	5.7%	0.96
Fari — eldercare	ETS	4.1%	1.00



Humanoid: shape, not level

Bass curve fits Goldman's anchors exactly (20k/2025 → 250k/2030 → 1.4M/2035). But Morgan Stanley projects ~13M in service by 2035 — ~3× apart. That spread is why humanoid is a scenario, not a forecast.

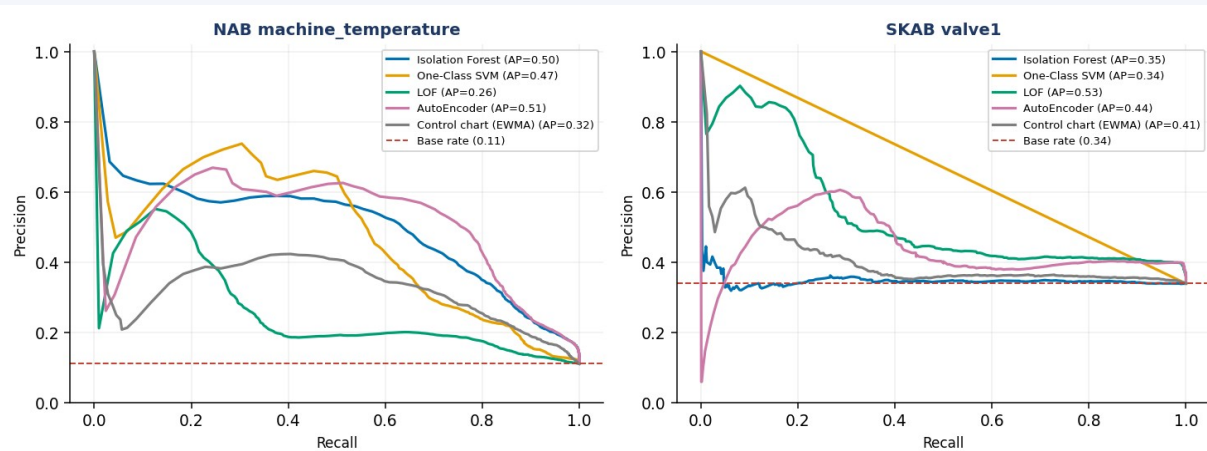
Sentinel: you can't threshold your way out of model quality

Week 11

Real public benchmarks (NAB — MIT; SKAB — GPL-3.0). No InGen data.

At 80% recall, the false-alarm rate bottoms out near 13% — whatever threshold we pick.

That ceiling is set by the model's discriminative power ($AP \approx 0.51$), not by threshold policy. The stated 5% budget is unreachable at that recall — and we say so rather than quietly relaxing the bar.



5 detectors, identical splits

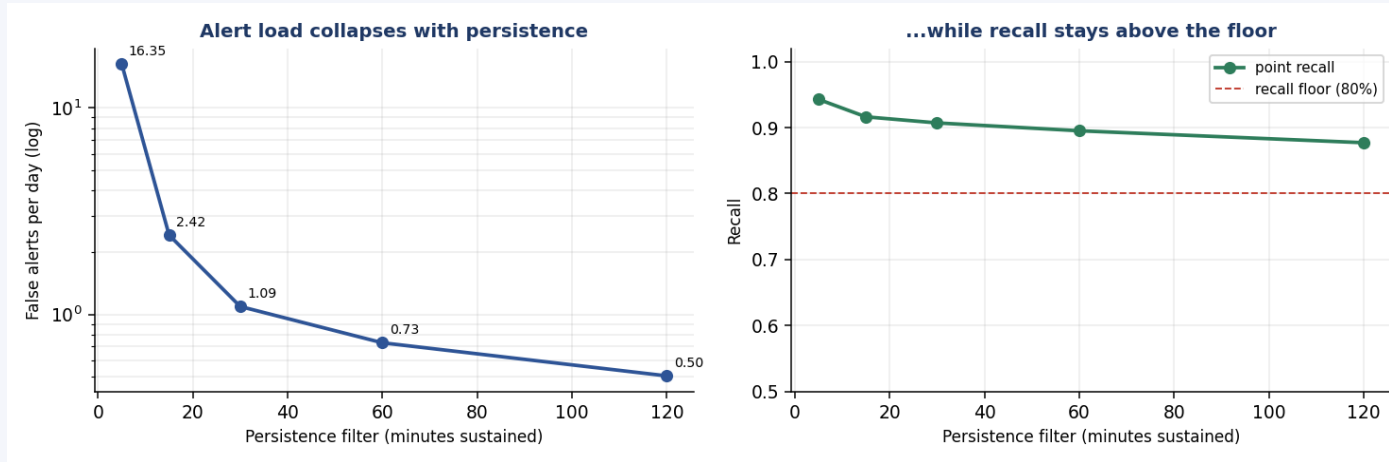
Isolation Forest · One-Class SVM · LOF · AutoEncoder (PyOD) — against an EWMA control chart, the thing a plant engineer would already do.

AutoEncoder leads NAB (F1 0.555), LOF leads SKAB (F1 0.545). Both clear the baseline — the bar that justifies the complexity.

Fitted unsupervised on a clean warm-up window — how a Sentinel unit would be baselined at commissioning.

...but persistence cuts alert load 33×

The point-level rate is the wrong metric. Operators experience alerts, not readings.



16.35

false alerts / day

no persistence filter

0.50

false alerts / day

120-min persistence

Recall RISES to 88%

All 4/4 real failures still caught. Real failures are sustained; false positives are isolated spikes.

Adaptive thresholds lost — and the reason is the lesson

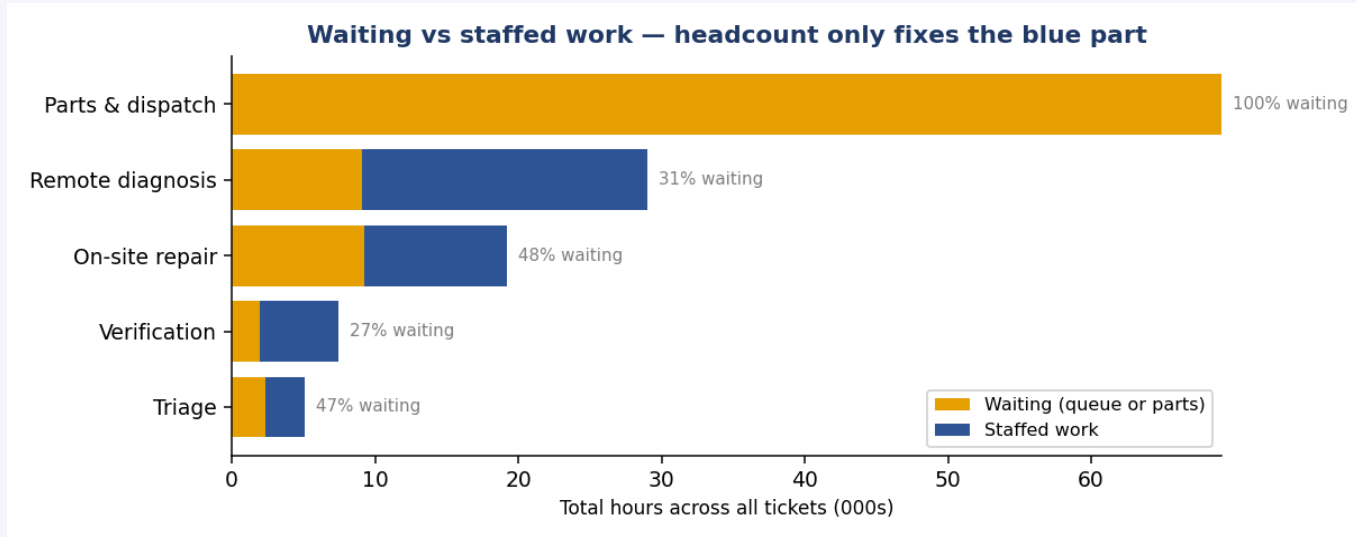
Rolling quantile cut false alarms (13.4% → 3.3%) but collapsed recall (80% → 33%). NAB's failures run ~2 days, so a rolling baseline absorbs the anomaly and lifts the threshold exactly when it should hold. Survived a full parameter sweep; a baseline-freeze test confirmed the diagnosis. Verdict: fixed + persistence, re-baselined on a schedule.

Recommendation: two tiers — fast (intrusion) + sustained (degradation). Thresholds transfer as a method, not as constants.

Support is a parts problem, not a people problem

Week 12

Where the fleet-support cycle time actually goes.



53%

of all process time is Parts & dispatch — and 100% of it is unstaffed waiting.

Nobody is working during it, so no amount of headcount shortens it. On-site repair, by contrast, is 48% queueing — which headcount can shorten. That distinction drives the whole recommendation.

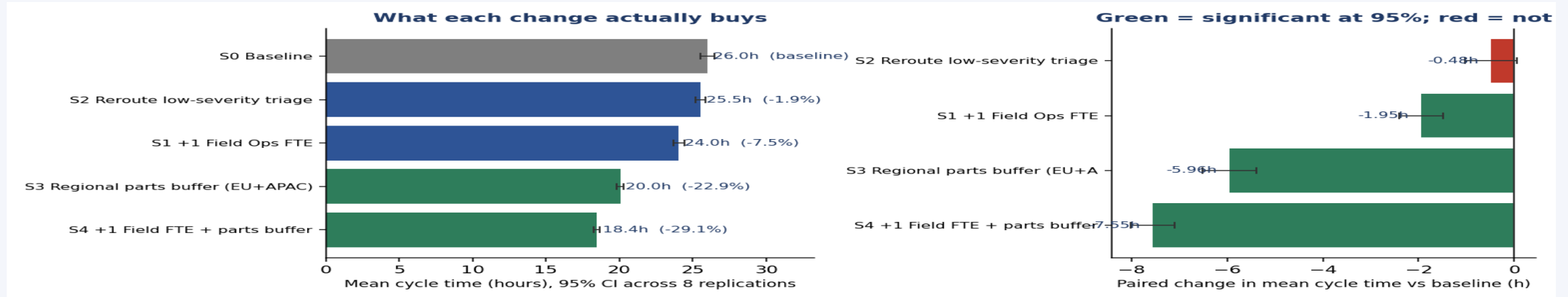
The drivers you'd reach for explain 4% of the variance. One structural fact explains 72%.

Product, severity, geography, weekday, team workload → $R^2 = 0.04$. Add "did this ticket need a physical part?" → $R^2 = 0.72$, and a dispatched ticket takes +578% longer [+555%, +601%]. Severity and queue length are real but second-order.

What each fix would actually buy

Week 12

8 replications per scenario, paired to identical seeds, 95% CIs.



-23%

Regional parts buffer

P90 falls 64h → 47h. The tail is what customers complain about.

-8%

+1 Field Ops FTE

Nearly zeroes field-queue wait (4.9h → 0.4h) — but that stage is only ~15% of the process.

-1.9%

Reroute Tier-1 triage

Not significant. Triage is 4% of process time. The obvious lever does nothing.

Amdahl's law in practice: perfect staffing of a 15% stage cannot buy more than 15%. Both changes together: -29%.

What public data could not answer

Stated plainly, because the caveats matter as much as the findings.

No internal InGen data — anywhere

Every figure is public or clearly-labelled synthetic. Market bottom-ups use assumed penetration, not observed conversion. These analyses are directional, not decision-grade.

The synthetic warehouse fixes shape, not fact

Weeks 7–9 and 12 rest on generated data. Nothing there measures InGen's actual fleet, support org or pipeline. Scorecard KPI targets are my assumptions.

Forecasts are momentum reads

A 0–100 search index answers "is attention rising?" — not "how many units will we sell?"

Week 12's numbers are model properties

The –23% is a simulation result, not a measurement. The method was validated (7/9 generating effects recovered); the number still needs real data.

Dashboards built, not published

Specs, extracts, prototypes and design log are complete. No live URLs exist, and none are claimed.

Left out on purpose

Competitor hiring counts (no dated snapshot), vendor deployment claims (marketing-sourced), frozen EV/Revenue multiples.

Week 11 is the exception that proves the point: real public benchmarks were reachable, and its results are measurements on real sensor data.

What I'd do next

1

Market Sentinel Prime AI harder

First on market attractiveness, last on demand signal — the widest gap in the portfolio.

2

Fund a regional parts buffer before hiring

~23% vs ~8%. Parts is 53% of process time and 100% waiting.

3

Ship two-tier alerting on Sentinel

Fast tier for intrusion; 120-min sustained tier at ~0.5 false alerts/day.

4

Treat humanoid as an option, not a plan

Credible analysts differ ~3× on 2035. Scenario-bound it.

The one ask

Read access to one small, anonymized slice of real InGen data.

The synthetic warehouse already matches the intended schema, so real data slots in with minimal rework — and it dissolves most of the limitations at once:

- the warehouse becomes real
- the dashboards become operational
- Week 12's diagnostic becomes a measurement
- the forecasts gain a calibration target

Weeks 1–13 complete · 13 self-contained modules, each with README, plan, status and tests · HANDOFF.md documents how to pick up every workstream · thank you.